Linux Server Workshop

Grab your piece of paper. You'll need it soon.

What is a server?

A server is just a computer that is always on, always connected, waiting for other machines to talk to it.

Almost all servers run **Linux**, free, stable, runs without a screen or keyboard attached.

What can you do with a server?

- Host a website or web app
- Run a Discord bot
- Build and expose an API
- Host a game server (Minecraft, Valheim...)
- Store files (your personal google drive)
- Schedule automated tasks and scripts
- Run a database
- Run Docker containers

Where do you get one?

You rent one. A company owns physical machines in a data centre and you pay for a slice of it. This is called a **VPS** (Virtual Private Server).

Hetzner

Probably the best value VPS provider in Europe. Starts around **€5/month.**

	VCPU	RAM	SSD	Price	
CPX11	2 AMD	2 GB	40 GB	max. **€ 5.99** /month	**€ 0.0096** /hour
CPX21	3 AMD	4 GB	80 GB	max. **€ 9.99** /month	**€ 0.0160** /hour
CPX31	4 AMD	8 GB	160 GB	max. **€ 17.99** /month	**€ 0.0288** /hour
CPX41	8 AMD	16 GB	240 GB	max. **€ 32.99** /month	**€ 0.0529** /hour
CPX51	16 AMD	32 GB	360 GB	max. **€ 71.49** /month	**€ 0.1146** /hour
CPX12	1 AMD	2 GB	40 GB	max. **€ 8.49** /month	**€ 0.0136** /hour

Contabo

Cheaper than Hetzner for the specs you get. Less stable but totally fine.

Good pick if you want a lot of RAM or disk on a budget.

Cloud VPS 10	Cloud VPS 20 (Best Selling VPS)	Cloud VPS 30	Cloud VPS 40
~~€4.50~~	~~€7.00~~	~~€14.00~~	~~€25.00~~
3.60 €/ month	5.60 €/ month	11.20 €/ month	20.00 €/ month
for a 12-month term	for a 12-month term	for a 12-month term	for a 12-month term
Get Started	Get Started	Get Started	Get Started
4 vCPU Cores	6 vCPU Cores	8 vCPU Cores	12 vCPU Cores
8 GB RAM	12 GB RAM	24 GB RAM	48 GB RAM
75 GB NVMe or 150 GB SSD	100 GB NVMe or 200 GB SSD	200 GB NVMe or 400 GB SSD	250 GB NVMe or 500 GB SSD
1 Snapshot	2 Snapshots	3 Snapshots	3 Snapshots
200 Mbit/s Port	300 Mbit/s Port	600 Mbit/s Port	800 Mbit/s Port
	No Setup Fee	No Setup Fee	No Setup Fee

You as students also have access to strato

https://strato-new.claaudia.aau.dk/auth/login/?next=/

> These servers have weird network setup, but you can access them outside of uni net with the right network rules.

What we're using today - Please follow along

Servers provided by **SupercomputerClub** running on the university infrastructure.

You were handed a piece of paper with an IP and a password. That's all you need.

> These servers have **private IPs** - only reachable on the university network. Relevant when we talk about domains later.

What is SSH?

SSH lets you open a terminal on a remote machine over the network. When you type a command, it runs on the server instead of your laptop.

```
$ ssh ubuntu@172.25.11.238
ubuntu@172.25.11.238's password:

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-106-generic x86_64)

  System load:  0.0                Processes:       129
  Usage of /:   13.3% of 18.33GB   Users logged in: 0
  Memory usage: 9%                 IPv4 address:    172.25.11.238

ubuntu@vm238:~$
```

That `$` prompt means you're in. Everything you type now runs on the server.

Connecting - Waiting for everyone to connect

SSH comes installed on every platform.

Mac / Linux - open Terminal:

```
ssh ubuntu@YOUR_IP_ADDRESS
```

Windows 11 - open the Terminal app (search in Start menu):

```
ssh ubuntu@YOUR_IP_ADDRESS
```

Windows 10 - use Command Prompt (cmd):

```
ssh ubuntu@YOUR_IP_ADDRESS
```

Type `yes` if asked about the fingerprint, then enter the password from your paper.

Task 01

Host a webpage

Follow along - everyone does this one together.

Step 1 - SSH in and get the repo

Connect to your server using the IP and password from your paper.

Once you're in, clone the repo:

```
git clone https://github.com/DeeKahy/Linux-Server-Workshop.git
cd Linux-Server-Workshop
```

Step 2 - Go into the project folder

```
cd 01-web-server
ls
```

You'll see three files:

```
README.md    index.html    server.py
```

`index.html` is the page. `server.py` is what serves it.

Step 3 - Start the server

```
python3 server.py
```

```
Serving on http://0.0.0.0:8080
Visit http://YOUR_SERVER_IP:8080 in your browser
Press ctrl+c to stop
```

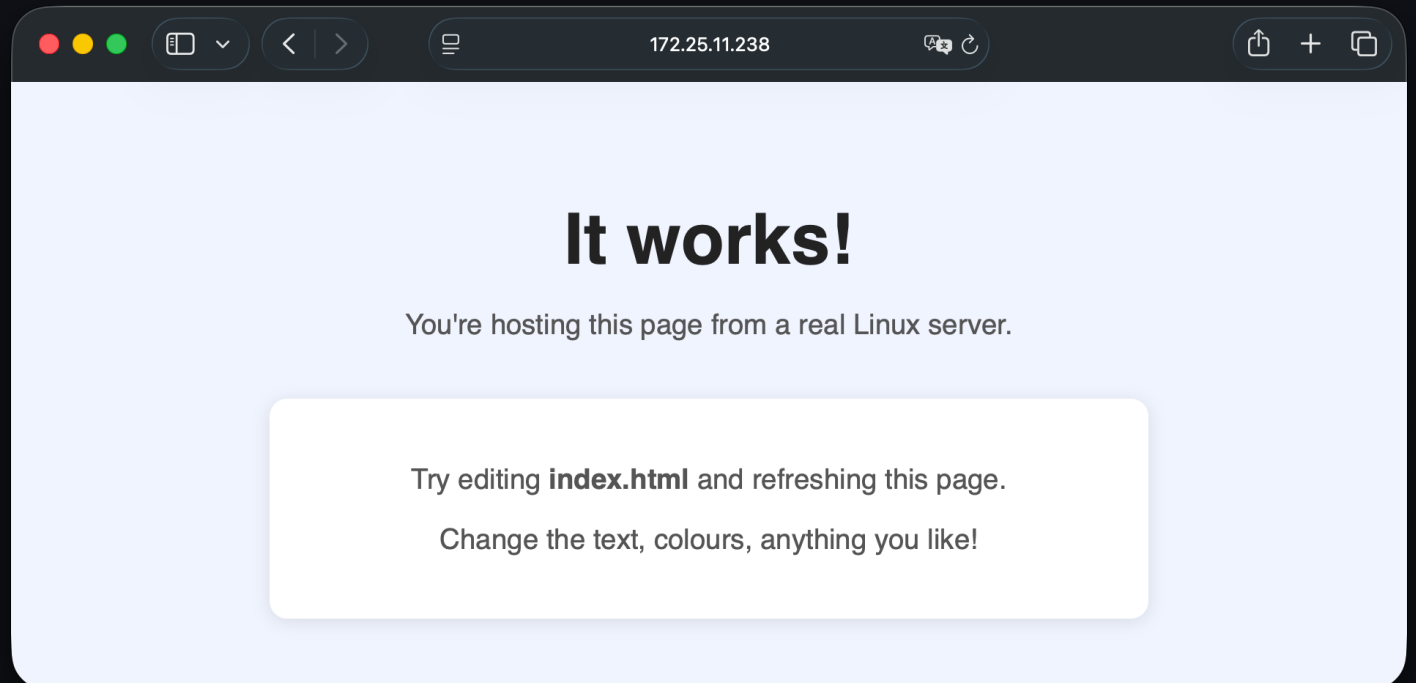
Leave this running and open a browser on your laptop.

Step 4 - Visit it in your browser

Go to:

```
http://YOUR_SERVER_IP:8080
```

⚠️Make sure it's **http://** not https:// - the server doesn't have a certificate so HTTPS won't work.

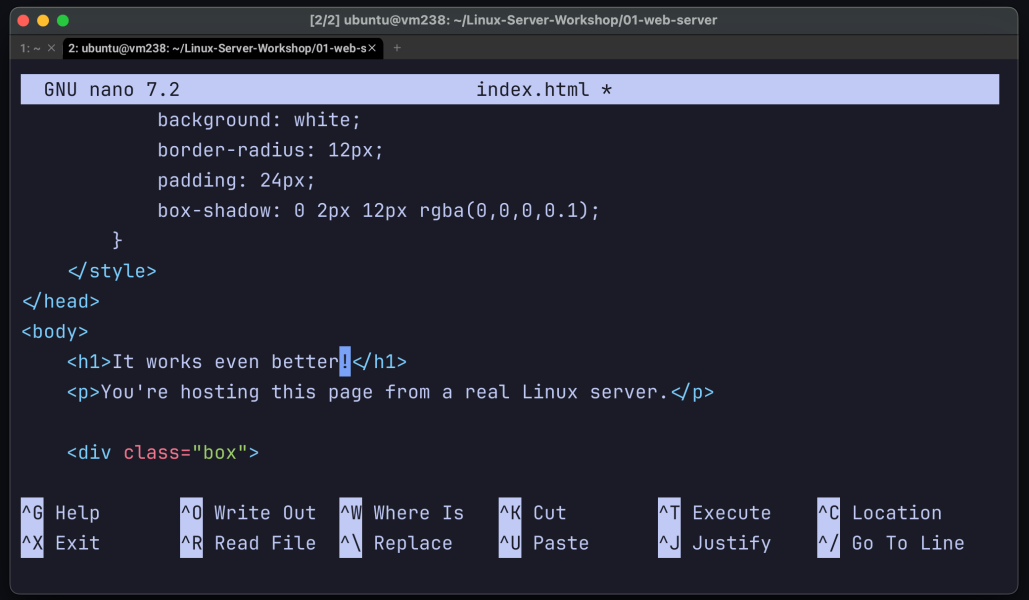


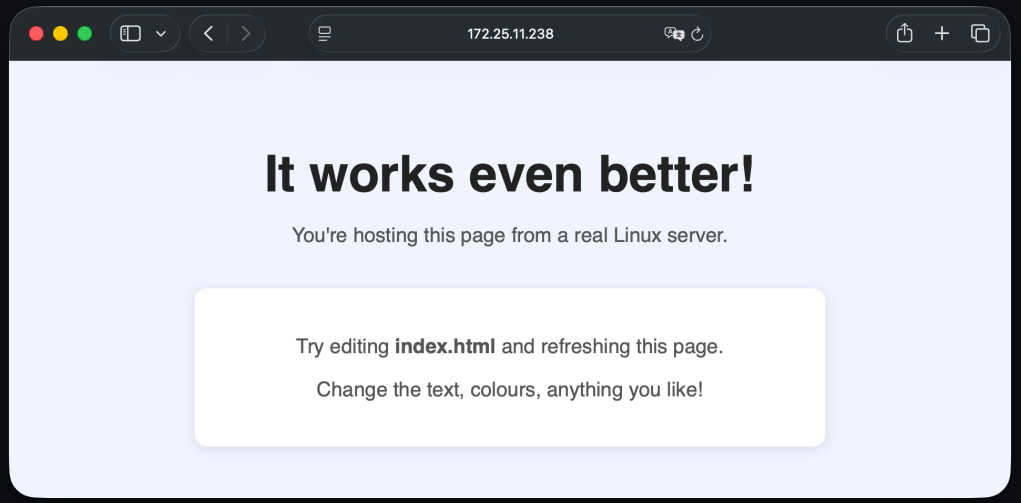
Step 5 - Edit the page

Stop the server with `ctrl+c`, then open the HTML file:

```
nano index.html
```

Change the heading or any text. Save: `ctrl+o` then `ctrl+x`. Start the server again and refresh.





Why we can't do DNS today

Our servers have **private IPs** - they only exist inside the university network. The internet has no way to route to them.

To run a public-facing server with a domain you need:

1. A VPS with a **public IP** - Hetzner or Contabo
2. A **domain** - buy through some free provider or Cloudflare (at cost)
3. Add an **A record** pointing the domain at your IP

Worth trying at home this weekend if you're curious.

Tasks - explore at your own pace

Task	What
01 - Web server	Done - you just did this
02 - Discord bot	Ping/pong bot, needs a token
03 - systemd service	Run programs as background services
04 - Flask API	Build a REST API with Python
05 - Bash scripting	Write shell scripts
06 - Cron jobs	Schedule tasks
07 - UFW firewall	**Control which ports are open**
08 - SSH keys	**Log in without a password**
09 - Auto updates	Keep the server patched
10 - TCP chat	Group chat over the local network

Go explore

Open any task folder and read its `README.md`.

Ask for help any time.

Task 10 (TCP chat) is a group activity - find us when you want to run it.